

ASSOCIATION AND CAUSATION

When considering the observed association with two variables one must not assume a causal relationship between them. The association between the two variables could be due to either coincidence or the presence of a third (or more) variable known as a **lurking variable** or **confounding factor** or variable. Consider the following situations.

Situation 1

If we found that there was a high correlation between eating in restaurants and high cholesterol we could not conclude that eating in restaurants causes one to have high cholesterol. This high association may be due to either coincidence or the presence of another variable or variables. There are many other variables in this equation that we did not take into account that may be the cause or help contribute to the cause, such as alcohol intake, type of food eaten, age of people in the survey, stress levels of people involved and so on.

The value of r cannot be the basis on which we can state that one variable is the cause of another. One must take into account the fact that most cause-effect relationships involve more than just the two variables being considered also the two variables in question may not be genuinely related, they are spuriously related and may be caused by some third variable which may or may not be obvious.

Situation 2

If a high correlation was found between the number of ice creams sold and the number of drowning deaths could we conclude that an increase in ice cream sales causes an increase in drownings or the greater the number of drownings the greater the sale of ice creams? Not a logical conclusion.

The variables ice cream sales and the number of drownings are related to a common variable, the daily temperature. The higher the temperature the more ice creams are sold and also the more people go swimming thus the more drownings. The lurking variable here is the daily temperature.

Situation 3

Over recent years sales of mobile phones and sales of athletic footwear have increased significantly and hence there is a high correlation between sales of mobile phones and sales of athletic footwear.

Just because there is a high correlation between these two variables we cannot assume that buying mobile phones causes people to buy athletic footwear or buying athletic footwear causes people to buy mobile phones.

In this case the high association between sales of mobile phones and sales of athletic footwear may be due to **coincidence**. Alternatively it may be due to higher degree of affluence or some other variable (s).

Situation 4

A newspaper reported that students who eat breakfast perform better at school. Follow up studies found that there was a correlation between eating breakfast and higher academic achievement, also the researchers found that eating breakfast did not affect the intelligence of students. The research found that the type of student who did not eat breakfast due to various circumstances in their daily lives were also the type of student that had trouble at school.

It may appear to some people that the newspaper reported on a causation when it really reported on a correlation or association.

Summing up: Just because there is a high value of r this does not mean that one variable causes the other. There could be a lurking variable(s) that actually is the cause.

EXERCISE 2C

1. The scattergram below shows the relationship between the two variables x and y .

- (a) For the data shown in the scattergram find,
(i) mean of x (ii) mean of y

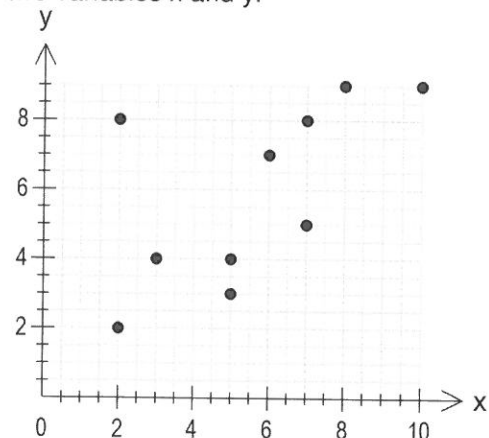
- (iii) standard deviation of x

- (iv) standard deviation of y

- (v) correlation coefficient

- (b) The point (2, 8) was deemed to be too far from the trend of the data and it was removed.

Recalculate the statistics in (a) after the removal of the point (2, 8).



- (c) Comment on the effect the removal of the outlier had on the correlation coefficient.

2. The table below gives the number of hours 10 students spent watching videos during the weekend prior to their science test and their results for that test out of a possible mark of 10.

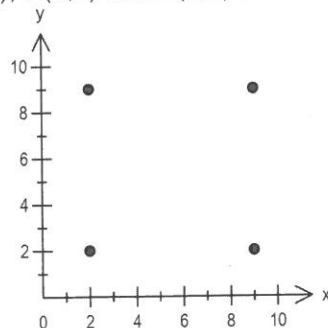
Hours spent watching videos	2	3	4	4	5	6	7	8	9	10
Science test mark	9	9	7	6	8	5	4	5	4	9

- (a) Identify the response and explanatory variables. Justify your choice.
- (b) Find the mean test mark and the mean number of hours spent watching videos.
- (c) Find the median test mark.
- (d) Is there a relationship between the number of hours spent watching videos and the marks achieved in the Science test? Justify your answer mathematically.

On checking the test papers it was found that the mark of 9 for the student who had watched 10 hours of videos should have been a mark of 2.

- (e) Find the correct mean test mark and median test mark.
- (f) Does this correction alter the relationship between test marks and hours spent watching videos? Mathematically justify your answer.

3. The four points A(2,9), B(2,2), C(9,2) and D(9,9) shown on the scatter diagram form a square.



Answer the following questions **without the aid of a calculator** and check your answers using a calculator.

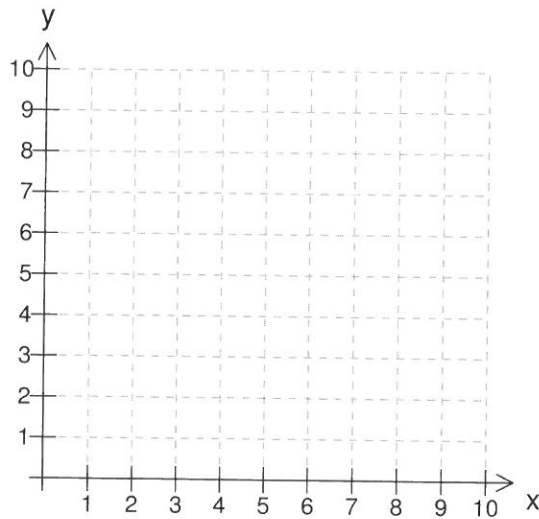
- (a) By carefully considering the relationship between the points write down correlation coefficient for the four points.
- (b) Write down the correlation coefficient if the points A and C are removed.
- (c) Write down the correlation coefficient if the points B and D are removed.
- (d) Find r_{xy} if A and B are removed.
- (e) Find r_{xy} if A and D are removed.

If the point A only is removed the correlation coefficient of the remaining points is 0.5.

- (f) If the point B only is removed, what is the value of r_{xy} ?
- (g) If the point C only is removed, what is the value of r_{xy} ?
- (h) If the point representing the centre of the square was introduced, what would the value of r_{xy} be?

4. (a) Sketch a scatter diagram of the following set of scores.

x	1	2	3	4	4	5	6	7	8	9
y	3	4	3	4	8	4	5	6	6	7

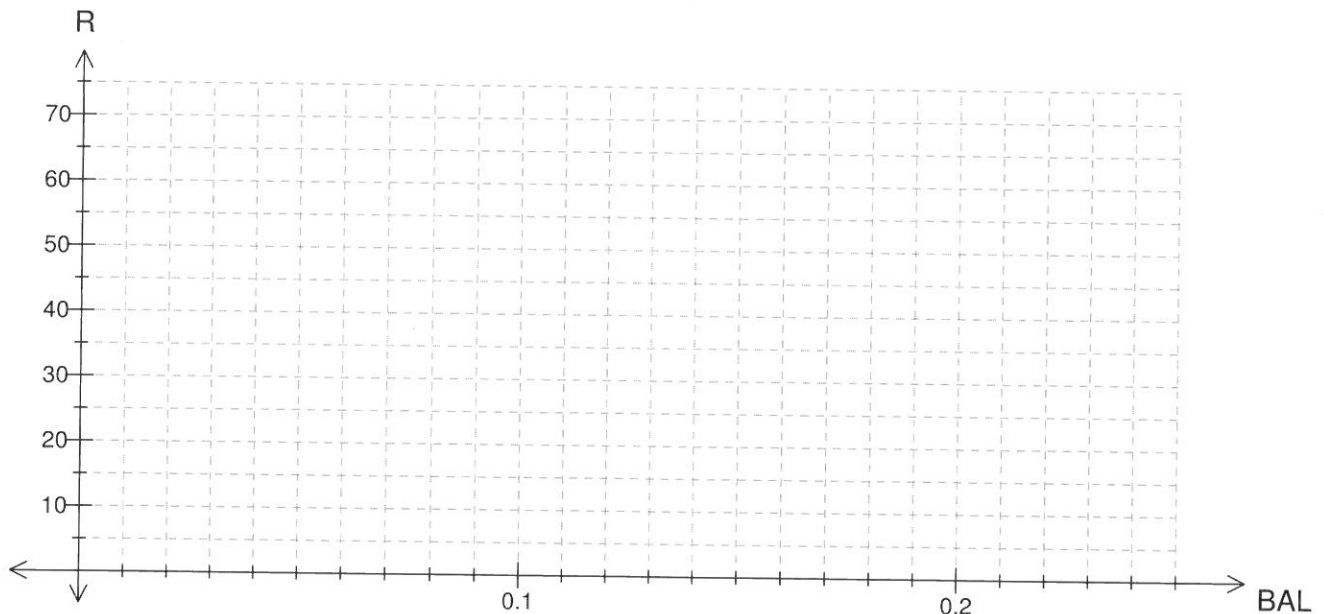


- (b) Comment on the relationship between the variables x and y . Justify mathematically.
- (c) It was found that the point $(4, 8)$ should have been registered as $(8, 4)$. Comment on the corrected relationship between the variables x and y . Justify mathematically.
- (d) Compare and comment on your results with (a) and (b) above.

5. The risk of a motor vehicle accident is associated with the driver's blood alcohol level (BAL). The risk factor R is the number of times more likely it is that a driver who has alcohol in their bloodstream will have an crash than a driver with a blood alcohol level of zero. The table below shows the relationship between blood alcohol level and R , the risk factor.

Blood Alcohol Level	0	0.04	0.08	0.10	0.15	0.17	0.18	0.20
Risk Factor R	1	2	4	15	28	40	55	70

- (a) On the grid below draw the scatter plot for the given data.



(b) Calculate the value of r , the correlation coefficient, for the given data set.

(c) Describe the data using your r value.

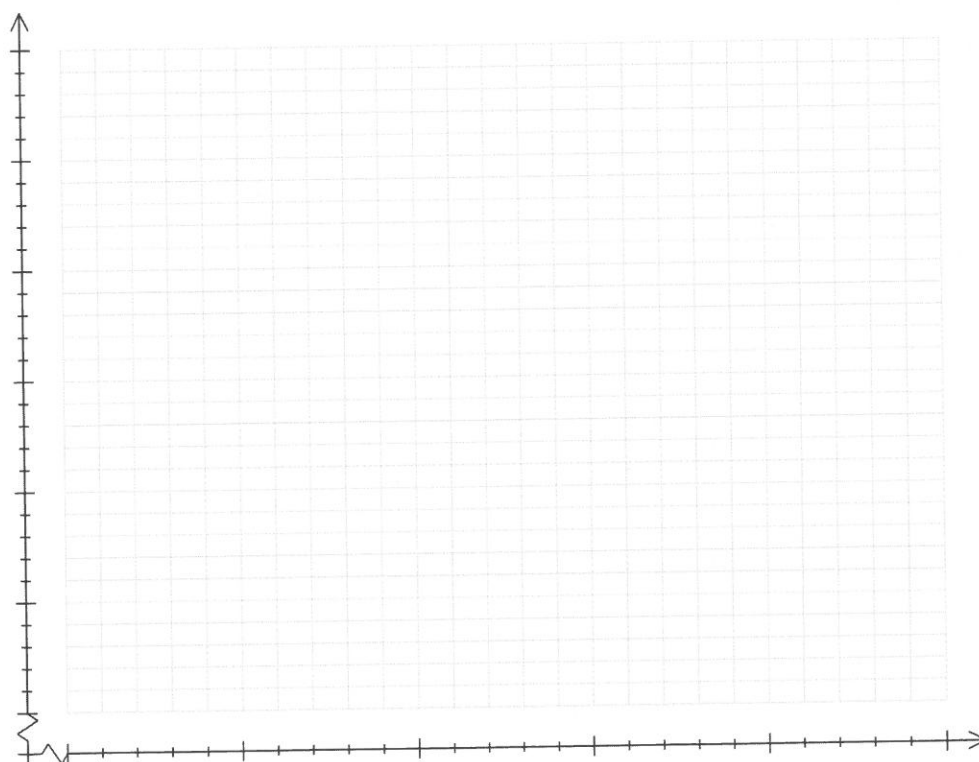
(d) Explain why the correlation coefficient is not useful in describing the given data set.

6. A group of 12 year old school children were given two fitness tests. The first test was a measure of maximum lung pressure before exercise and the second was a measure of heart rate taken one minute after strenuous exercise. The test results have been tabled below.

Child number	1	2	3	4	5	6	7	8	9	10	11	12	13
Lung pressure	48	26	45	36	40	43	30	29	38	44	35	39	33
Heart rate	80	95	84	90	85	84	82	94	90	82	88	86	92

(a) Is there mathematical evidence of an association between maximum lung pressure and recovery heart rate? Explain using a suitable statistic.

(b) On the give grid construct a scatter plot for the given data.



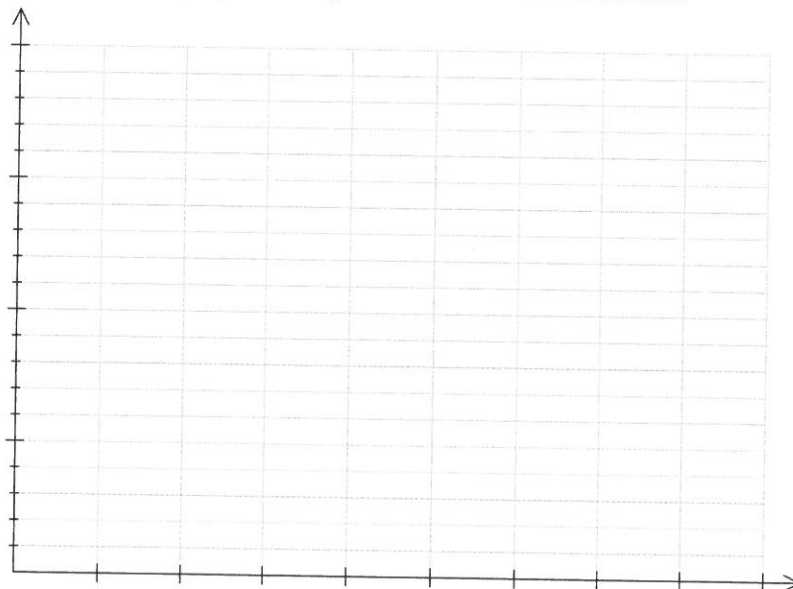
(c) From your scatter plot identify the child with the greatest variation from the main group.

(d) Would it make much difference to the correlation coefficient if the data from this child were treated as an outlier and removed? Discuss.

7. The table shows the number of lifeguards on duty and the number of rescues of near drownings on Sundays during the summer of 2016 collated by Jodie a budding statistician.

Number of lifeguards on duty	1	4	3	5	2	5	6	4	7	3	4	2
Number of rescues of near-drownings	3	9	6	11	6	14	18	12	19	8	10	5

(a) Construct a scatter graph of the given data on the axes below.

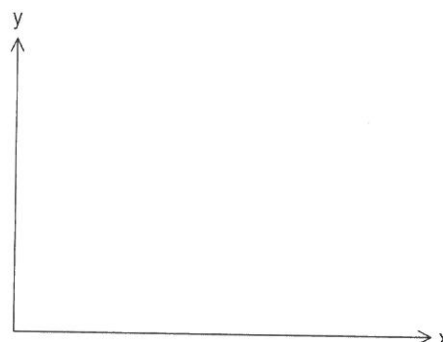


(b) Comment on the relationship between the number of lifeguards on duty and the number of rescues of near drownings.

(c) Jodie calculated the value of the correlation coefficient for the given data and found it to be 0.96 rounded to two decimal places and concluded that the more lifeguards on duty caused more near drownings and hence it was probably better to have no lifeguards on duty. Discuss Jodie's conclusions.

8. Recent research showed that there was a high correlation between the number of mobile phones per household and the life expectancy per person among many countries. Does this mean that mobile phones cause people to live longer? Discuss

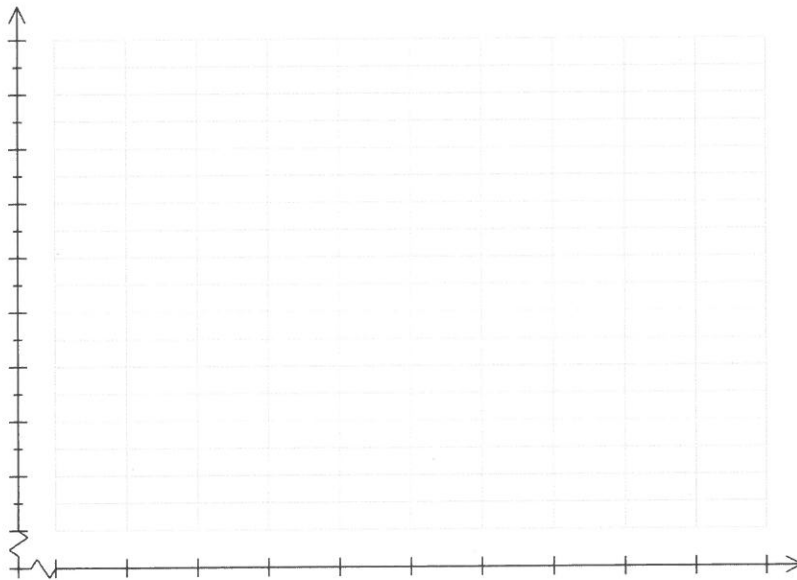
9. A correlation of $r = 0$ indicates no linear relationship between the two given variables. But, this does not mean that there is no relationship between the two variables. Sketch a scatter plot in which there is a strong relationship between the variables, but the correlation would be 0 or near $r = 0$.



10. A study was conducted by a group of mathematics students to determine whether there was a relationship between performance and the cost of graphical calculators. The calculators were subjected to a number of tests and were given a rating expressed as a percentage. The group tested ten different graphical calculators and their findings are tabled below.

Calculator type	Performance (P%)	Cost (\$C)
A	60	150
B	78	200
C	96	210
D	62	170
E	57	140
F	67	185
G	88	175
H	55	205
I	75	160
J	90	210

- (a) Construct a scatter graph of the given data on the axes below.

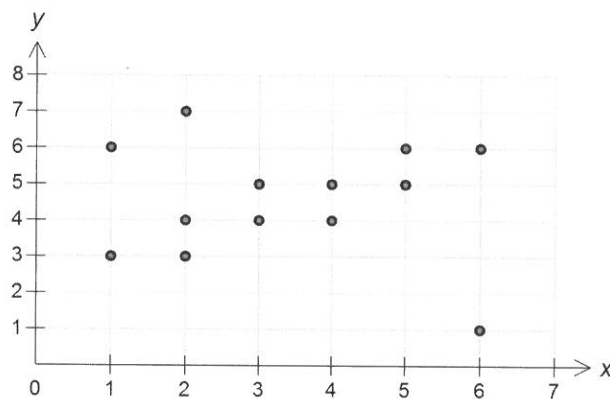


- (b) Comment on the relationship between performance and the cost of these calculators.
- (c) Calculate the correlation coefficient for this data set and describe the linear relationship that the value of r represents.
- (d) Do any of the calculator types qualify as a possible outlier(s)? If so, state which type(s) and justify your choice.
- (e) Remove the outlier(s) and recalculate the correlation coefficient.
- (f) Comment on your answer to (e).
- (g) In writing up a report on their findings one student stated the following:
 'high performance of a calculator means that cost will be high'
 Comment on this student's statement.

11. A survey of a large number of cities revealed a high correlation between the number of police officers and the number of crimes committed.
Does this mean that more police officers are causing more crimes to be committed? Discuss.

12. A study revealed a high correlation between the number of television sets per household and the life expectancy per person among many countries. Can we conclude therefore that television sets cause people to live longer? Comment.

13. The scattergram shows the relationship between the variables x and y .



- (a) Calculate the value of the correlation coefficient and comment on the relationship between x and y .

Closer examination of the scattergram reveals that three data points do not appear to fit the data set.

- (b) List the outliers for this data set.
- (c) Crop the data set to exclude these outliers and recalculate the correlation coefficient. Comment on the cropped relationship.
- (d) If the outliers were real values of the data set of the distribution, justify their removal.